Shape of data

1

a Symmetrical **b** Negative skew **c** Negative skew **d** Positive skew

e Negative skew **f** Positive skew **g** Symmetrical **h** Symmetrical

i Negative skew **j** Positive skew

2

a Negative skew **b** **i** 6 **ii** 8 **c** 2 **d** No

3

a Positive skew **b** **i** 2 **ii** 4 **c** 2 **d** Yes, 9

4

a 18 years old **b** 8 years

c 36 years **d** 21.74 years old

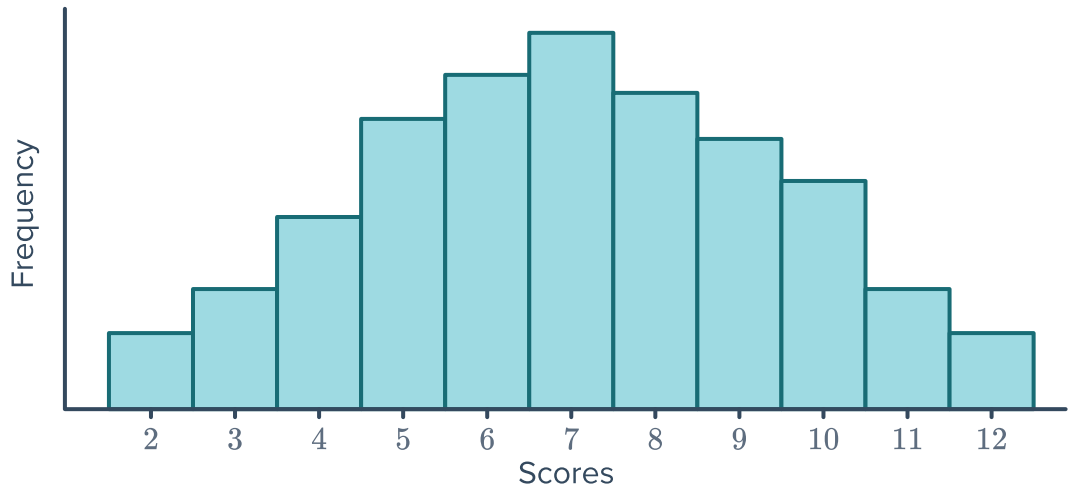
e Positively skewed

5 The mean is greater than 70.

6

a **i** 3 **ii** 6 **iii** 3

b



7

a **A** On each roll, each number is equally likely to appear.

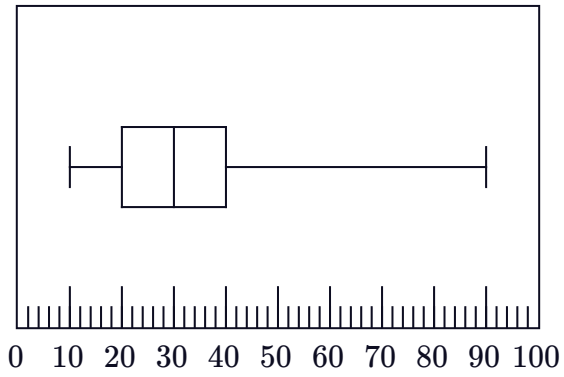
b Uniformly distributed

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

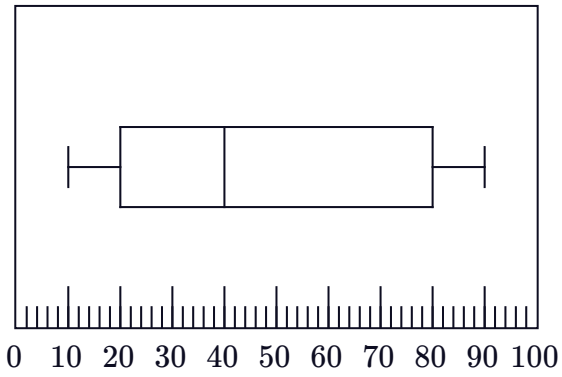
Box plots and histograms

8

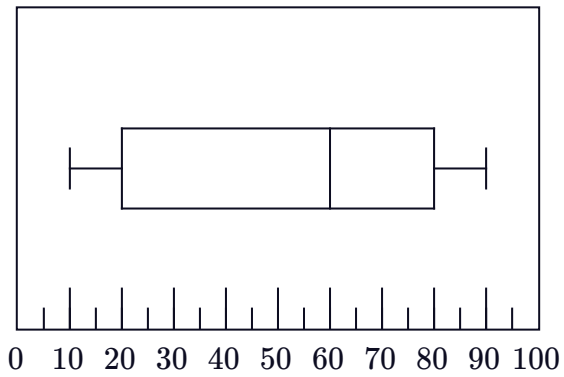
a



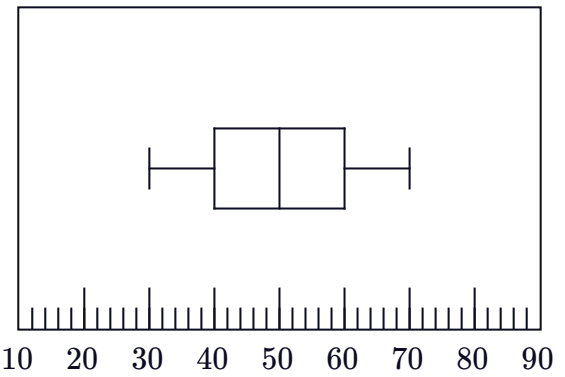
b



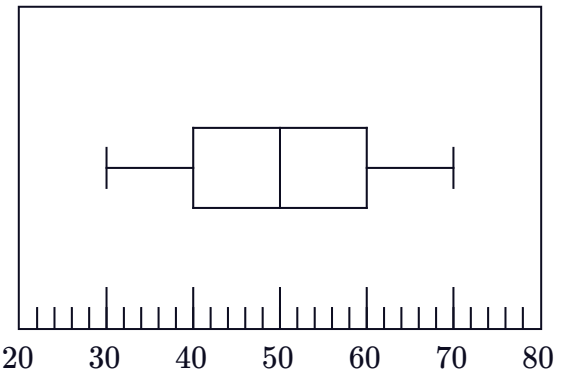
c



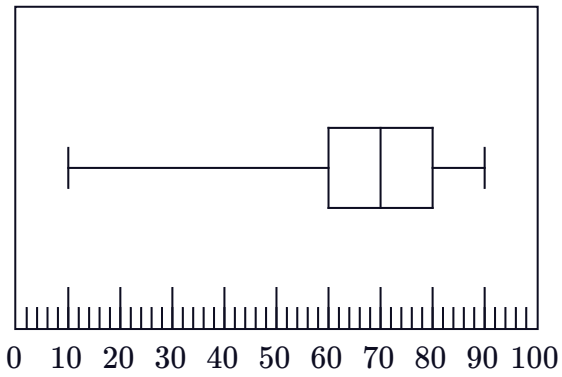
d



e



f



9

- Box Plot 1/Histogram A
- Box Plot 2/Histogram B
- Box Plot 3/Histogram D
- Box Plot 4/Histogram C

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10

a Yes **b** No **c** No **d** Yes

e Yes **f** No

11

a The box plot has a long tail to the right which indicates positive skew, while the histogram does not appear to be skewed. The median for the histogram is roughly in the middle, while the median of the box plot is located further to the left.

b The box plot indicates that the data is widely spread, while the histogram indicates that most of the data occurs near the median.

Measure of center

12

a Mean **b** Mode **c** Median

13

a Median **b** Mean **c** Mean **d** Mean

14

a $\$816\,000$

b $\$760\,000$

c $\$725\,000$

d $\$760\,000$

e Median, because the outlier does not affect it.

15

a 12.5 kg

b 14.2 kg

c 13.875 kg

d 14.25 kg

e Median, because it was the least affected by the outlier.

16 The mean, median and mode are all equal for this data set, so any would give the same representation.

17 Median because it is less affected by the three unusually high scores, and the mode in this case does not represent the center of the data well, it is the least score and has only one more employee than several other categories.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18 She should use the mean as it is the greater measure of average as it is affected by the outlier.

Boxplots and outliers

19

a

i 11 **ii** -10.5 **iii** 33.5

b

i 10 **ii** -9 **iii** 31

20

a

i

Minimum	1
Q1	2
Median	4
Q3	6
Maximum	9

ii 4 **iii** -4 **iv** 12

v Yes **vi** Yes

b

i

Minimum	2
Q1	3
Median	6
Q3	9
Maximum	10

ii 6 **iii** -6 **iv** 18

v No **vi** No

c

i

Minimum	1
Q1	5
Median	7
Q3	10
Maximum	12

ii 5 **iii** -2.5 **iv** 17.5

v Yes **vi** No

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

21

a

i

Minimum	1
Q1	2
Median	5
Q3	8
Maximum	10

ii No

iii No

b

i

Minimum	4
Q1	6
Median	9
Q3	10
Maximum	11

ii Yes

iii No

22 $n = 24$

23 $n = 21$

24

a 30

b 42.5

c 26.5

d Positively skewed

e 76

f They do none to moderate amounts of exercise.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647